

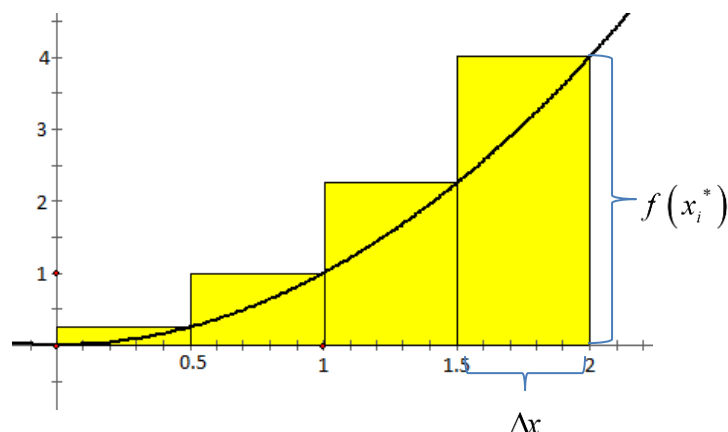
1.1 Areas Between Curves

A Quick Review:

Example: Let A be the area of the region that lies above the x -axis and under the graph of $f(x) = x^2$ between $x = 0$ & $x = 2$. Estimate the area by using a Right Riemann Sum with 4 rectangles.

$$\begin{aligned} A &\approx 0.5[f(0.5) + f(1) + f(1.5) + f(2)] \\ &= 0.5[0.25 + 1 + 2.25 + 4] \\ &= 0.5[7.5] \\ &= 3.75 \end{aligned}$$

This is an overestimate of the area.



Definition: If f is a continuous function on $[a, b]$, then the definite integral of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

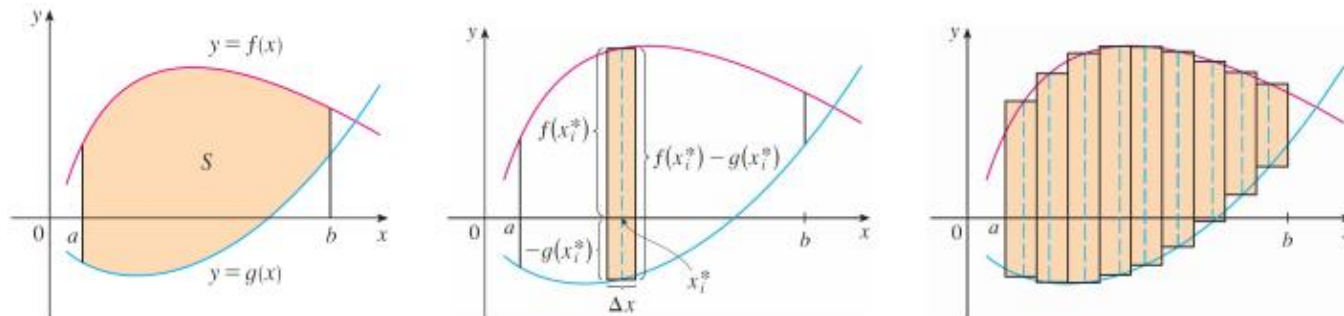
Annotations for the formula:

- Let the number of rectangles go to ∞ (points to $\lim_{n \rightarrow \infty}$)
- Add up the areas of n rectangles (points to $\sum_{i=1}^n$)
- Height of each rectangle (points to $f(x_i^*)$)
- Width of each rectangle (points to Δx)

provided that this limit exists and gives the same value for all possible choices of sample points x_i^* . If it does exist, we say that f is **integrable** on $[a, b]$.

Areas Between Two Curves:

General Idea:



Consider the picture below that has the region S which is determined by the area between the functions $f(x)$ and $g(x)$ on the interval $[a, b]$. To find this area we will use a Riemann sum by dividing the region into n intervals (which correspond to n rectangles) of equal width. So we have

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n [f(x_i^*) - g(x_i^*)] \Delta x$$

Where x_i^* is a sample point from the i th interval. We can recognize this limit as the definite integral of $f - g$.

Definition (Area Between Two Curves w/respect to x): The area A of the region bounded by the curves $y = f(x)$ & $y = g(x)$, and the lines $x = a$ & $x = b$, where f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, is

$$A = \int_a^b [f(x) - g(x)] dx$$

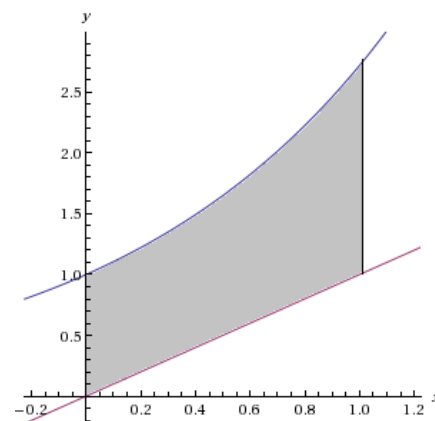
Example 1: Find the area of the region bounded by $y = e^x$, $y = x$, $x = 0$, & $x = 1$.

Setup:

$$A = \int_0^1 (e^x - x) dx$$

Evaluate:

$$A = \int_0^1 (e^x - x) dx = \left(e^x - \frac{1}{2}x^2 \right) \Big|_0^1 = \left[e - \frac{1}{2} \right] - [1 - 0] = e - \frac{3}{2}$$



Example 2: Find the area of the region bounded by the curves $y = \sin(x)$, $y = \cos(x)$, $x = 0$, & $x = \frac{\pi}{2}$.

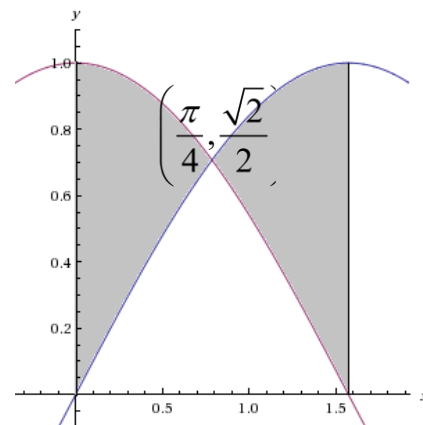
When we look at the picture we see that the “top” function changes after the intersection. We must split this problem up into two different integrals. To do this we need to find the point of intersection.

We know that $\sin(x) = \cos(x) \Leftrightarrow x = \frac{\pi}{4}$ on the given interval for x .

Therefore we write that:

Setup:

$$A = \int_0^{\pi/4} [\cos(x) - \sin(x)] dx + \int_{\pi/4}^{\pi/2} [\sin(x) - \cos(x)] dx$$



Evaluate:

$$\begin{aligned} A &= \int_0^{\pi/4} [\cos(x) - \sin(x)] dx + \int_{\pi/4}^{\pi/2} [\sin(x) - \cos(x)] dx \\ &= [\sin(x) + \cos(x)] \Big|_0^{\pi/4} + [-\cos(x) - \sin(x)] \Big|_{\pi/4}^{\pi/2} \\ &= \left[\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right] - [0 + 1] + [0 - 1] - \left[-\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} \right] \\ &= \sqrt{2} - 1 - 1 + \sqrt{2} \\ &= 2\sqrt{2} - 2 \end{aligned}$$

Example 3: Find the area enclosed by the curves $y = x - 1$ & $y^2 = 2x + 6$.

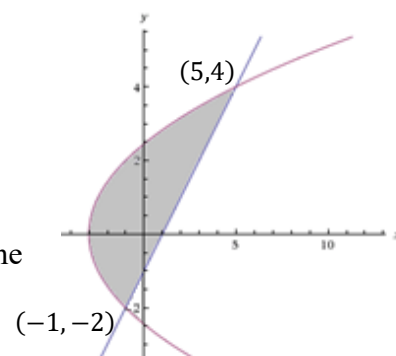
We start by finding the intersection points of the region. We see that

$$\begin{aligned} y + 1 &= \frac{1}{2}y^2 - 3 \Leftrightarrow 2y + 2 = y^2 - 6 \Leftrightarrow y^2 - 2y - 8 = 0 \\ &\Leftrightarrow (y - 4)(y + 2) = 0 \Leftrightarrow y = 4 \text{ or } y = -2 \end{aligned}$$

If $y = 4$, then $x = 5$ and we have the point $(5, 4)$.

If $y = -2$, then $x = -1$ and we have the point $(-1, -2)$.

We notice that on the interval $[-1, 5]$ the parabola is the *top* function and the line is the *bottom*, however on the interval $[-3, -1]$ the parabola is *both* the top & bottom function.

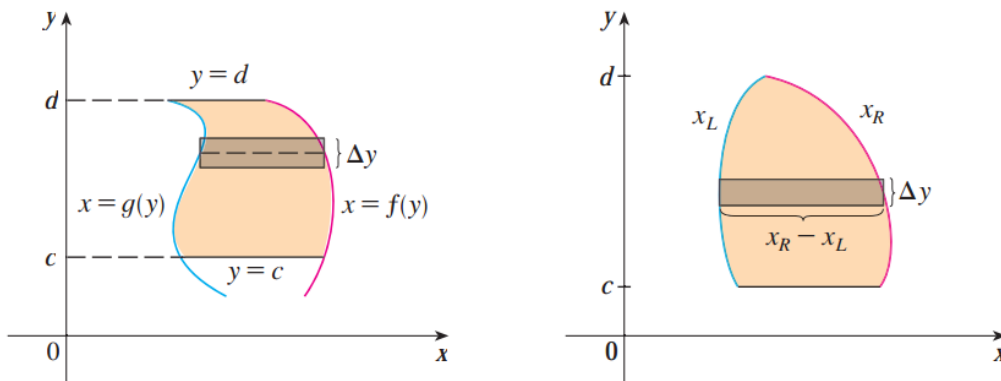


At times like these it may be helpful to write the parabola as two separate functions of x or to integrate with respect to y

Integrating with Respect to y (instead of x)

General Idea:

We should note that not all curves can be regarded as y as a function of x . Some curves are best treated as $x = f(y)$, where x is a function of y . If we consider the picture below we can see that we can still use the idea of a Riemann sum.



Definition (Area Between Two Curves with respect to y): The area A of the region bounded by the curves $x = f(y)$ & $x = g(y)$, and the lines $y = a$ & $y = b$, where f and g are continuous and $f(y) \geq g(y)$ for all y in $[a, b]$, is

$$A = \int_a^b [f(y) - g(y)] dy$$

Example 3 (continued): Find the area enclosed by the curves $y = x - 1$ & $y^2 = 2x + 6$.

We begin by solve our equations for x so that we can view these curves as functions of y .

$$y = x - 1 \Rightarrow x = y + 1$$

$$y^2 = 2x + 6 \Rightarrow x = \frac{1}{2}y^2 - 3$$

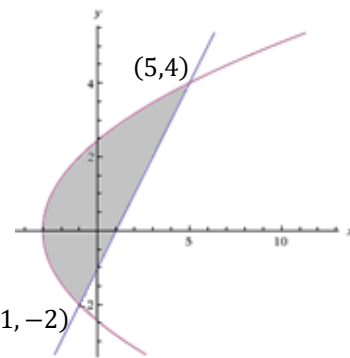
We see that the linear function forms the right side of the region and the parabola function forms the left side of the region. Therefore we write the following.

Setup:

$$A = \int_{-2}^4 \left[(y+1) - \left(\frac{1}{2}y^2 - 3 \right) \right] dy$$

Evaluate:

$$\begin{aligned} A &= \int_{-2}^4 \left[(y+1) - \left(\frac{1}{2}y^2 - 3 \right) \right] dy = \int_{-2}^4 \left[-\frac{1}{2}y^2 + y + 4 \right] dy = \left[-\frac{1}{6}y^3 + \frac{1}{2}y^2 + 4y \right]_{-2}^4 \\ &= \left[-\frac{1}{6}(64) + \frac{1}{2}(16) + 16 \right] - \left[-\frac{1}{6}(-8) + \frac{1}{2}(4) - 8 \right] = \left[\frac{40}{3} \right] - \left[-\frac{14}{3} \right] = 18 \end{aligned}$$



Example 4: Find the area enclosed by the curves $y = \sqrt[4]{x}$ & $y = \sqrt{2-x}$ and the x -axis.

a) Find this area by integrating with respect to y .

We see that the region has a right boundary of $y = \sqrt{2-x}$ and a left boundary of $y = \sqrt[4]{x}$. We must solve these equations for x .

$$y = \sqrt[4]{x} \Rightarrow x = y^4$$

$$y = \sqrt{2-x} \Rightarrow x = 2 - y^2$$

We also find the intersection points of these curves.

$$\begin{aligned} y^4 = 2 - y^2 &\Leftrightarrow y^4 + y^2 - 2 = 0 \Leftrightarrow (y^2 + 2)(y^2 - 1) = 0 \\ &\Leftrightarrow y^2 = -2 \text{ or } y^2 = 1 \Leftrightarrow y = \pm 1 \end{aligned}$$

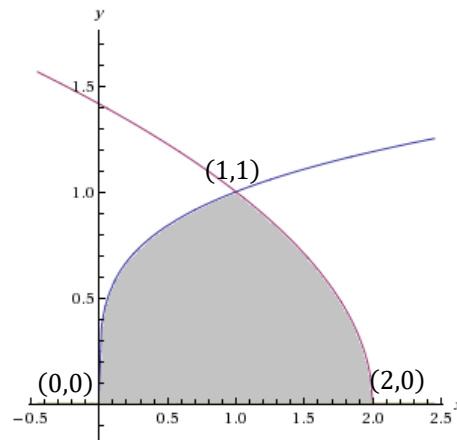
So clearly we have an intersection point at $(1,1)$. Therefore we have:

Setup:

$$A = \int_0^1 (2 - y^2 - y^4) dy$$

Evaluate:

$$A = \int_0^1 (2 - y^2 - y^4) dy = \left(2y - \frac{1}{3}y^3 - \frac{1}{5}y^5 \right) \Big|_0^1 = \left[2 - \frac{1}{3} - \frac{1}{5} \right] - [0] = \frac{22}{15}$$



b) Find this area by integrating with respect to x .

This approach will require 2 integrals since *top* boundary changes when $x = 1$. The bottom boundary is always $y = 0$.

Setup:

$$A = \int_0^1 (\sqrt[4]{x} - 0) dx + \int_1^2 (\sqrt{2-x} - 0) dx$$

Evaluate:

$$A = \int_0^1 (\sqrt[4]{x} - 0) dx + \int_1^2 (\sqrt{2-x} - 0) dx = \int_0^1 \sqrt[4]{x} dx + \int_1^2 \sqrt{2-x} dx$$

Let $u = 2 - x$ and thus $du = -dx \Leftrightarrow -du = dx$.

Our limits of integration become: $u(1) = 2 - (1) = 1$ & $u(2) = 2 - 2 = 0$. So we have:

$$A = \int_0^1 \sqrt[4]{x} dx + \int_1^2 \sqrt{2-x} dx = \int_0^1 x^{1/4} dx - \int_1^0 \sqrt{u} du = \frac{4}{5} x^{5/4} \Big|_0^1 - \frac{2}{3} u^{3/2} \Big|_1^0 = \left[\frac{4}{5} - 0 \right] - \left[0 - \frac{2}{3} \right] = \frac{4}{5} + \frac{2}{3} = \frac{22}{15}$$

